

Envelope Theorem and Duality

ECON 320 - Introduction to Mathematical Economics

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Learning Objectives

- Understand the envelope theorem for constrained and unconstrained problems.
- Apply the theorem in consumer, producer, and profit-maximization settings.
- Develop intuition for duality in economic optimization.
- Learn the relationship between Lagrange multipliers, shadow prices, and value functions.

1. Introduction

Many economic optimization problems depend on an external parameter (price, income, technology). We frequently want to know how the *optimal value* of the objective changes when the parameter changes, without re-solving the optimization problem completely.

The **Envelope Theorem** provides an elegant method to compute this derivative. The related concept of **Duality** expresses an optimization problem in an equivalent but economically insightful form.

2. The Optimization Setup

Consider the constrained maximization problem:

$$\max_{x_1, \dots, x_n} f(x_1, \dots, x_n, a)$$

subject to

$$g(x_1, \dots, x_n, a) = 0,$$

where a is a parameter.

Let $x_i^*(a)$ denote the optimal choices, and let the **value function** be

$$V(a) = f(x_1^*(a), \dots, x_n^*(a), a).$$

3. The Envelope Theorem

Theorem 1 (Envelope Theorem — Constrained Case). *For the constrained maximization problem above, the derivative of the value function is:*

$$\frac{dV}{da} = \frac{\partial f}{\partial a}(x^*, a) - \lambda \frac{\partial g}{\partial a}(x^*, a)$$

where λ is the optimal Lagrange multiplier.

Key idea: At the optimum, the indirect effects through dx_i^*/da cancel out by the first-order conditions.

Unconstrained Version

For:

$$\max_x f(x, a), \quad f_x(x^*, a) = 0,$$

$$\frac{dV}{da} = \frac{\partial f}{\partial a}(x^*(a), a).$$

4. Economic Interpretation of the Envelope Theorem

Interpretation: Only the *direct effect* of the parameter on the objective and constraint matters. All indirect effects through optimal choices vanish.

Examples:

- In consumer theory: $\frac{dU^*}{dM} = \lambda$ = marginal utility of income.
- In profit maximization: $\frac{\partial \pi^*}{\partial w} = -L^*$ = negative of optimal labor usage.
- In cost minimization: $\frac{dC}{dQ} = \lambda$ = marginal cost.

5. Applications

5.1 Consumer Utility Maximization

$$\max_{x,y} U(x, y) \quad \text{s.t.} \quad p_x x + p_y y = M.$$

Lagrangian:

$$\mathcal{L} = U(x, y) + \lambda(M - p_x x - p_y y).$$

Envelope Theorem:

$$\frac{dU^*}{dM} = \lambda.$$

Thus λ is the **marginal utility of income**.

5.2 Profit Maximization

$$\pi = \max_{K,L} [pF(K, L) - wL - rK].$$

Envelope Theorem gives:

$$\frac{\partial \pi}{\partial w} = -L^*, \quad \frac{\partial \pi}{\partial r} = -K^*, \quad \frac{\partial \pi}{\partial p} = F(K^*, L^*).$$

Thus:

- Profit falls with each input price at the rate of input usage.
- Profit rises with output price at the rate of output.

5.3 Cost Minimization

$$C(w, r, Q) = \min_{K,L} [wL + rK] \quad \text{s.t.} \quad F(K, L) = Q.$$

Envelope Theorem:

$$\frac{\partial C}{\partial Q} = \lambda = \text{marginal cost}.$$

6. Duality

Definition 1. ***Duality** refers to the correspondence between an optimization problem (the primal) and another optimization problem (the dual) that characterizes the same economic behavior in a different form.*

6.1 Consumer Duality

Problem	Objective	Constraint
Utility Maximization	Maximize $U(x, y)$	$p_x x + p_y y = M$
Expenditure Minimization	Minimize $p_x x + p_y y$	$U(x, y) = \bar{U}$

Both yield the same tangency condition:

$$MRS = \frac{p_x}{p_y}.$$

6.2 Producer Duality

Problem	Objective	Constraint
Cost Minimization	$\min C = wL + rK$	$F(K, L) = Q$
Profit Maximization	$\max \pi = pF(K, L) - wL - rK$	None

Key identity:

$$\pi(p, w, r) = pQ - C(w, r, Q).$$

7. Shephard's Lemma

For the cost function:

$$C(w, r, Q) = \min_{K, L} \{wL + rK : F(K, L) = Q\},$$

$$\boxed{\frac{\partial C}{\partial w} = L^*, \quad \frac{\partial C}{\partial r} = K^*}$$

That is, the derivative of the cost function with respect to input prices returns the **conditional factor demands**.

This is a direct application of the Envelope Theorem.

8. Graphical Insight

- The value function forms an **envelope** of the objective curves for different x .
- Each curve touches the envelope at the optimum.
- The envelope's slope gives the derivative of optimal value with respect to a parameter.

9. Summary of Key Results

$$\frac{dV}{da} = \frac{\partial f}{\partial a} - \lambda \frac{\partial g}{\partial a}$$

λ = shadow value of the constraint

$$\frac{\partial C}{\partial Q} = \text{marginal cost}$$

$$\frac{\partial \pi}{\partial w} = -L^*, \quad \frac{\partial \pi}{\partial r} = -K^*$$

$$\frac{\partial C}{\partial w} = L^*, \quad \frac{\partial C}{\partial r} = K^*$$

Envelope Theorem + Duality form one of the most powerful toolkits for modern microeconomic analysis.